INCH-POUND
MIL-P-23469/4B(SH)
19 February 1988
SUPERSEDING
MIL-P-23469/4A(SHIPS)
31 May 1967

MILITARY SPECIFICATION SHEET

Ⓑ PIN-RIVET, GROOVED, ROUND HEAD; STRAIGHT SHANK, MULTIPLE LOCKING GROOVES, ALUMINUM ALLOY, CORROSION-RESISTANT AND CARBON STEELS

Ⓑ This specification sheet is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

Ⓑ The requirements for acquiring the pins described herein shall consist of this specification and the latest issue of MIL-P-23469.

Ⓑ Round head, swage-locking pins shall be furnished in the following type and classes. Pins manufactured to this specification are intended for use with collars manufactured in accordance with MIL-P-23469.

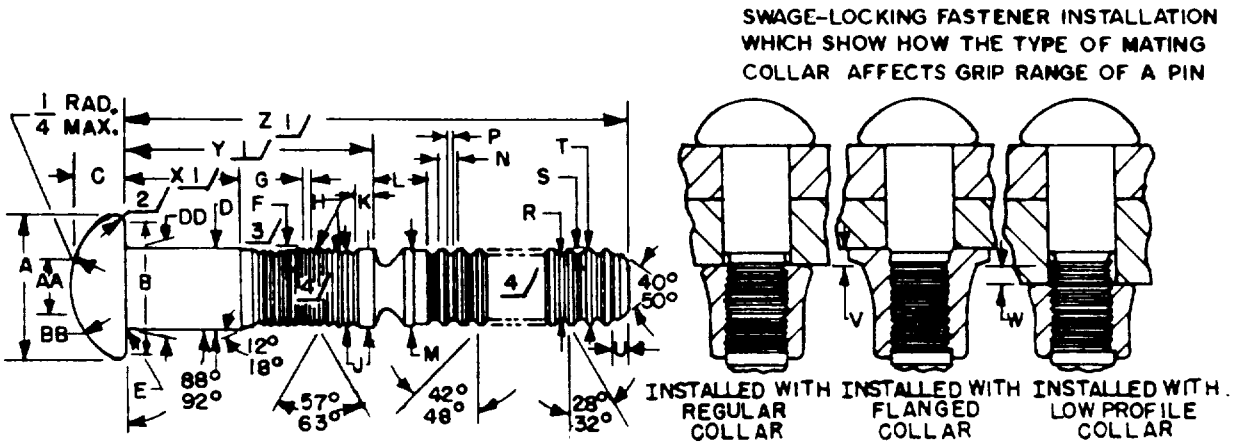
Type II - Multiple locking grooves (see figure 1)

Class 1 - Aluminum alloy
Class 2 - Corrosion resistant steel
Class 3 - Carbon steel

Ⓑ Denotes changes.

AMSC N/A FSC 5320
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SH 13203039

Nom dia	A Dia	AA Dia	B 5/	BB Ref Rad	C	CC 6/	D Dia	DD Dia 7/	E Rad	F	G Pitch	H Rad	J Dia	K Max	L Max
1/2	.938 .875	.360 .328	.828	.500	.302 .330	.030	.515 .493	.530	.021 .031	.025 .028	.039 .045	.002 .010	.495 .485	.094	.375
5/8	1.187 1.094	.485 .453	1.062	.609	.378 .415	.040	.642 .617	.675	.047 .062	.031 .035	.052 .058	.003 .013	.618 .608	.115	.375
3/4	1.438 1.328	.532 .500	1.250	.734	.455 .500	.047	.768 .741	.800	.047 .062	.036 .040	.059 .065	.003 .015	.740 .730	.135	.500
* 7/8	1.656 1.547	.641 .609	1.437	.844	.531 .578	.055	.895 .866	.938	.047 .062	.043 .048	.068 .076	.003 .017	.870 .860	.156	.625
* 1	1.906 1.781	.766 .734	1.625	.969	.591 .640	.062	1.022 .990	1.063	.078 .093	.052 .058	.080 .088	.004 .020	.993 .983	.180	.625

Nom dia	M Dia	N 3/ Pitch	P	R Dia	S	T Max dia	U Min	V 9/ Flanged collar	W Low 10/ profile collar	Hole preparation data: Size drill	Hole preparation data: Max hole dia
1/2	.473 .449	.0615 .0635	.016 .018	.427 .423	.020 .023	.473	.062	.125	.140	17/32 (.531)	9/16
5/8	.615 .570	.0771 .0791	.020 .023	.530 .524	.026 .030	.590	.078	.156	.180	21/32 (.656)	11/16
3/4	.740 .682	.0927 .0947	.025 .028	.637 .631	.032 .037	.711	.094	.188	.215	25/32 (.781)	13/16
* 7/8	.860 .790	.1084 .1104	.029 .032	.746 .738	.036 .041	.828	.109	.218	.250	29/32 (.906)	15/16
* 1	.980 .900	.124 .126	.032 .036	.857 .849	.041 .047	.951	.125	.250	.285	1-1/32 (1.031)	1-1/16

(B) * 7/8 and 1 inch not available in ALUM and CRES.

See footnotes at top of next page.

FIGURE 1. Type II - multiple locking grooves.

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1/ "X", "Y", and "Z" shown in table I.
2/ Radius is natural flow of material.
3/ Initial locking groove formed at juncture of angle and "D" body diameter.
4/ Dimensions indicated on locking groove and pull grooves are typical.
5/ Minimum bearing diameter.
6/ Conical surface of head to "D" diameter shall be concentric within "CC" TIR.
7/ Maximum seam diameter allowable.
8/ Accumulated pitch tolerance not to exceed plus or minus 0.002 on any 10 consecutive pull grooves.

Collar: Regular height - use grip shown in table I.

9/ Flanged - add flange thickness "V" to actual thickness to be fastened to determine "grip" to be used in specifying dash number, table I.
10/ Low profile - subtract "W" from actual thickness to be fastened to determine "grip" to be used in specifying dash number, table I.

NOTES:
1. Dimensions are in inches.
2. Shank must be straight within 0.004 TIR per inch of length.

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TABLE I. Multiple locking grooves.

Dash number identification															
M23469/4 1/			1/2 diameter					M23469/4 1/			3/4 diameter				
ALUM	CRES	Carbon steel	Grip range		X +.016 −.031	Y ±.020	Z +.00 −.26	ALUM	CRES	Carbon steel	Grip range		X +.016 −.031	Y ±.020	Z +.00 −.26
Class 1	Class 2	Class 3	Min	Max				Class 1	Class 2	Class 3	Min	Max			
-11604	-21604	-31604	.251	.500	.188	1.213	3.36	-12404	-22404	-32404	.251	.500	.188	1.518	4.34
-11608	-21608	-31608	.501	.750	.438	1.463	3.61	-12408	-22408	-32408	.501	.750	.438	1.768	4.59
-11612	-21612	-31612	.751	1.000	.688	1.713	3.86	-12412	-22412	-32412	.751	1.000	.688	2.018	4.84
-11616	-21616	-31616	1.001	1.250	.938	1.963	4.11	-12416	-22416	-32416	1.001	1.250	.938	2.268	5.09
-11620	-21620	-31620	1.251	1.500	1.188	2.213	4.36	-12420	-22420	-32420	1.251	1.500	1.188	2.518	5.34
-11624	-21624	-31624	1.501	1.750	1.438	2.463	4.61	-12424	-22424	-32424	1.501	1.750	1.438	2.768	5.59
-11628	-21628	-31628	1.751	2.000	1.688	2.713	4.86	-12428	-22428	-32428	1.751	2.000	1.688	3.018	5.84
-11632	-21632	-31632	2.001	2.250	1.938	2.963	5.11	-12432	-22432	-32432	2.001	2.250	1.938	3.268	6.09
-11636	-21636	-31636	2.251	2.500	2.188	3.213	5.36	-12436	-22436	-32436	2.251	2.500	2.188	3.518	6.34
-11640	-21640	-31640	2.501	2.750	2.438	3.463	5.61	-12440	-22440	-32440	2.501	2.750	2.438	3.768	6.59
-11644	-21644	-31644	2.751	3.000	2.688	3.713	5.86	-12444	-22444	-32444	2.751	3.000	2.688	4.018	6.84
-11648	-21648	-31648	3.001	3.250	2.938	3.963	6.11	-12448	-22448	-32448	3.001	3.250	2.938	4.268	7.09
-11652	-21652	-31652	3.251	3.500	3.188	4.213	6.36	-12452	-22452	-32452	3.251	3.500	3.188	4.518	7.34
-11656	-21656	-31656	3.501	3.750	3.438	4.463	6.61	-12456	-22456	-32456	3.501	3.750	3.438	4.768	7.59
-11660	-21660	-31660	3.751	4.000	3.688	4.713	6.86	-12460	-22460	-32460	3.751	4.000	3.688	5.018	7.84
-11664	-21664	-31664	4.001	4.250	3.938	4.963	7.11	-12464	-22464	-32464	4.001	4.250	3.938	5.268	8.09

See footnote at end of table.

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TABLE I. Multiple locking grooves. - Continued

M23469/4 1/			5/8 diameter					M23469/4 1/			7/8 diameter				
ALUM	CRES		Grip range		X +.016 −.031	Y ±.020	Z +.00 −.26			Carbon steel	Grip range		X +.016 −.031	Y ±.020	Z +.00 −.26
Class 1	Class 2		Min	Max						Class 3	Min	Max			
-12004	-22004		.251	.500	.188	1.363	3.78								
-12008	-22008		.501	.750	.438	1.613	4.03			-32808	.501	.750	.438	1.921	4.88
-12012	-22012		.751	1.000	.688	1.863	4.28			-32812	.751	1.000	.688	2.171	5.13
-12016	-22016		1.001	1.250	.938	2.113	4.53			-32816	1.001	1.250	.938	2.421	5.38
-12020	-22020		1.251	1.500	1.188	2.363	4.78			-32820	1.251	1.500	1.188	2.671	5.63
-12024	-22024		1.501	1.750	1.438	2.613	5.03			-32824	1.501	1.750	1.438	2.921	5.88
-12028	-22028		1.751	2.000	1.688	2.863	5.28			-32828	1.751	2.000	1.688	3.171	6.13
-12032	-22032		2.001	2.250	1.938	3.113	5.53			-32832	2.001	2.250	1.938	3.421	6.38
-12036	-22036		2.251	2.500	2.188	3.363	5.78			-32836	2.251	2.500	2.188	3.671	6.63
-12040	-22040		2.501	2.750	2.438	3.613	6.03			-32840	2.501	2.750	2.438	3.921	6.88
-12044	-22044		2.751	3.000	2.688	3.863	6.28			-32844	2.751	3.000	2.688	4.171	7.13
-12048	-22048		3.001	3.250	2.938	4.113	6.53			-32848	3.001	3.250	2.938	4.421	7.38
-12052	-22052		3.251	3.500	3.188	4.363	6.78			-32852	3.251	3.500	3.188	4.671	7.63
-12056	-22056		3.501	3.750	3.438	4.613	7.03			-32856	3.501	3.750	3.438	4.921	7.88
-12060	-22060		3.751	4.000	3.688	4.863	7.28			-32860	3.751	4.000	3.688	5.171	8.13
-12064	-22064		4.001	4.250	3.938	5.113	7.53			-32864	4.001	4.250	3.938	5.421	8.38

See footnote at end of table.

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TABLE I. Multiple locking grooves. - Continued

M23469/4 1/			5/8 diameter					M23469/4 1/			1" diameter				
		Carbon steel	Grip range		X +.016 -.031	Y ±.020	Z +.00 -.26			Carbon steel	Grip range		X +.016 -.031	Y ±.020	Z +.00 -.26
		Class 3	Min	Max						Class 3	Min	Max			
		-32004	.251	.500	.188	1.433	3.84								
		-32008	.501	.750	.438	1.683	4.09			-33208	.501	.750	.438	2.073	5.57
		-32012	.751	1.000	.688	1.933	4.34			-33212	.751	1.000	.688	2.323	5.82
		-32016	1.001	1.250	.938	2.183	4.59			-33216	1.001	1.250	.938	2.573	6.07
		-32020	1.251	1.500	1.188	2.433	4.84			-33220	1.251	1.500	1.188	2.823	6.32
		-32024	1.501	1.750	1.438	2.683	5.09			-33224	1.501	1.750	1.438	3.073	6.57
		-32028	1.751	2.000	1.688	2.933	5.34			-33228	1.751	2.000	1.688	3.323	6.82
		-32032	2.001	2.250	1.938	3.183	5.59			-33232	2.001	2.250	1.938	3.573	7.07
		-32036	2.251	2.500	2.188	3.433	5.84			-33236	2.251	2.500	2.188	3.823	7.32
		-32040	2.501	2.750	2.438	3.683	6.09			-33240	2.501	2.750	2.438	4.073	7.57
		-32044	2.751	3.000	2.688	3.933	6.34			-33244	2.751	3.000	2.688	4.323	7.82
		-32048	3.001	3.250	2.938	4.183	6.59			-33248	3.001	3.250	2.938	4.573	8.07
		-32052	3.251	3.500	3.188	4.433	6.84			-33252	3.251	3.500	3.188	4.823	8.32
		-32056	3.501	3.750	3.438	4.683	7.09			-33256	3.501	3.750	3.438	5.073	8.57
		-32060	3.751	4.000	3.688	4.933	7.34			-33260	3.751	4.000	3.688	5.323	8.82
		-32064	4.001	4.250	3.938	5.183	7.59			-33264	4.001	4.250	3.938	5.573	9.07

See footnote at top of next page.

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1/ Part number example - M23469/4-11614



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